

Sampling II: rejection and importance sampling

When p is known only up to a constant, draw from an easy q instead. Rejection keeps the draws that fall under $\tilde{p}$; importance sampling keeps them all and weights each by p/q .

THE QUESTION

A posterior is likelihood $\times$ prior over an unknown constant Z . How do we draw from it, or average over it, using only a distribution we can sample?

THE INTUITION

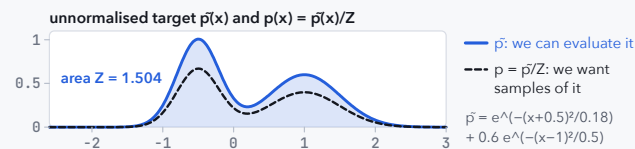
Draw from a proposal q . Rejection lifts q by M until Mq covers $\tilde{p}$ and keeps a draw with probability $\tilde{p}/(Mq)$. Importance sampling keeps every draw with weight $\tilde{p}/q$; the mean weight is Z .

KEY TAKEAWAY

Kept draws are exact samples, at rate Z/M . Weighted draws give $\sum_i \tilde{w}_i f(x_i) / \sum_i \tilde{w}_i$. Both need q to cover the tails of p , and both decay with dimension.

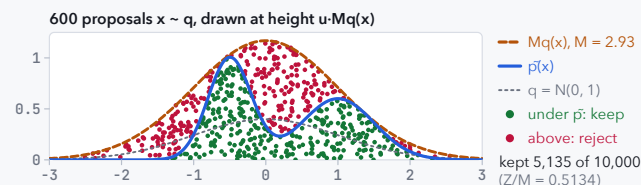
$\tilde{p} = \text{likelihood} \times \text{prior}$ $\rightarrow$ draw $x \sim q$ $\rightarrow$ keep if $u \cdot Mq(x) < \tilde{p}(x)$ $\rightarrow$ or weight $\tilde{w} = \tilde{p}/q$ $\rightarrow$ average; $\hat{Z} = \text{mean } \tilde{w}$

1 We can evaluate $\tilde{p}$ but cannot sample p



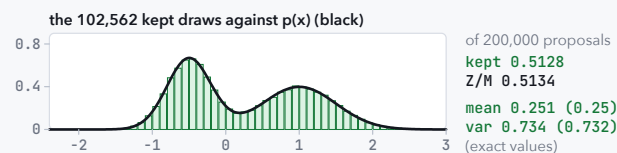
Monte Carlo wants $\mathbb{E}_p[f] \approx \frac{1}{N} \sum_i f(x_i)$ with $x_i \sim p$ (the Sampling I sheet). A posterior gives only $\tilde{p}(\theta) = p(\mathcal{D} | \theta)p(\theta)$; its constant $Z = \int \tilde{p} = p(\mathcal{D})$ is the evidence, and the inverse CDF of p is out of reach. Running example: the notebook's mixture $0.5 \mathcal{N}(-0.5, 0.3^2) + 0.5 \mathcal{N}(1, 0.5^2)$ without its constants, so $Z = 0.6\sqrt{2\pi} = 1.504$.

2 Rejection keeps the points under $\tilde{p}$



Pick q and M with $Mq(x) \geq \tilde{p}(x)$ for all x . Draw $x \sim q$ and $u \sim U(0, 1)$; keep x if $uMq(x) < \tilde{p}(x)$. The point $(x, uMq(x))$ is uniform under Mq , so the kept points are uniform under $\tilde{p}$ and their x follow p . Here $M = \max \tilde{p}/q = 2.929$ at $x = 1.33$, or $1.948Z$ (the notebook's grid gives 1.945).

3 Kept draws follow p exactly, at rate Z/M



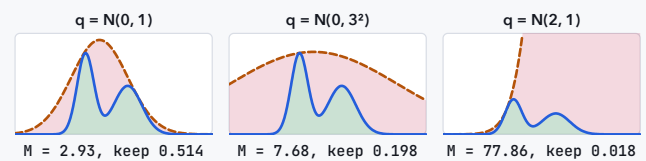
Bayes' rule with $P(a | x) = \tilde{p}(x)/(Mq(x))$ for acceptance a :

$$P(a) = \int P(a | x)q(x)dx = \frac{Z}{M}$$

$$p(x | a) = \frac{P(a | x)q(x)}{P(a)} = \frac{\tilde{p}(x)}{Z} = p(x)$$

The kept draws are independent and exact; only their count is random, $M/Z = 1.95$ proposals per draw here.

4 A loose envelope wastes draws



All area between Mq and $\tilde{p}$ is rejected, and M must hold at every x , so the worst spot sets it. The wide q wastes its tails; the shifted q must be lifted 78 times to cover the left bump. For $q = \mathcal{N}(0, \sigma_q^2)$ the rate peaks at 0.515 near $\sigma_q = 1$. Below $\sigma_q = 0.5$ no finite M exists: the $\mathcal{N}(1, 0.5^2)$ bump has heavier tails than q .

5 The keep rate falls as $(\sigma_p/\sigma_q)^D$



The slides' thought experiment: $\tilde{p} = \mathcal{N}(0, \sigma_p^2 \mathbf{I})$, $q = \mathcal{N}(0, \sigma_q^2 \mathbf{I})$, $\sigma_q = 1.1\sigma_p$. The ratio peaks at 0, so $M = 1.1^D$ and the rate is $1/M$: 0.91 at $D = 1$, 0.15 at 20, $7.3 \cdot 10^{-5}$ at 100. A q only 1% wider gives $4.8 \cdot 10^{-5}$ at $D = 1000$ (Bishop 2006, §11.1.2). Importance sampling on the same pair keeps $\text{ESS}/N = 0.985^D$ (beyond the slides); it decays too once q and p differ in shape.

6 Pause and think

Why not take M huge to be safe? The draws stay exact, but the rate Z/M falls.

Does rejection need Z ? No, only $\tilde{p}$ and a bound M . The kept fraction times M then estimates Z : $0.5135 \cdot 2.929 = 1.504$.

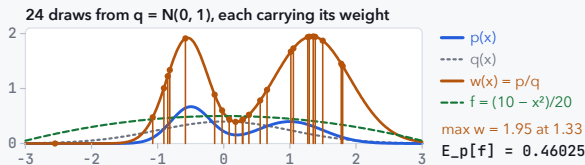
What if M is a little too small? The kept density becomes $\min(\tilde{p}, Mq)$, normalised. The notebook's grid M is 0.13% low and clips $2 \cdot 10^{-5}$ of the mass near $x = 1.33$.

What about 100 dimensions? A rate of $7 \cdot 10^{-5}$: weight the draws instead (page 2), or move locally (the MCMC sheet).

Weight every draw instead of rejecting it

The weight p/q corrects for sampling from the wrong distribution. Normalising the weights removes Z ; their average is Z .

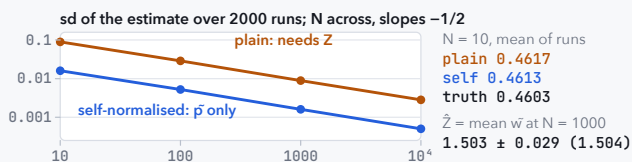
7 Importance sampling weights draws by p/q



$$\mathbb{E}_p[f] = \int f(x) \frac{p(x)}{q(x)} q(x) dx \approx \frac{1}{N} \sum_{i=1}^N f(x_i) w(x_i), \quad x_i \sim q$$

Unbiased whenever $q > 0$ where $f p \neq 0$. The variance is $\text{Var}_q(fw)/N$ in place of $\text{Var}_p(f)/N$: here 0.081 against 0.0021, since f is nearly flat and the weights are not. The notebook's $f = (10 - x^2)/20$ has $\mathbb{E}_p[f] = 0.46025$.

8 Self-normalised weights need only $\tilde{p}$

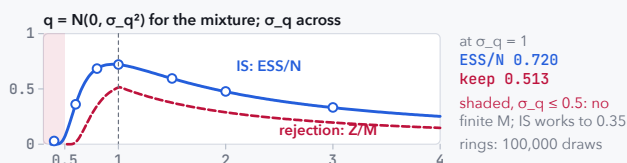


With $\tilde{w}_i = \tilde{p}(x_i)/q(x_i)$: $\hat{Z} = \frac{1}{N} \sum_i \tilde{w}_i$ is unbiased for Z , and

$$\hat{I} = \frac{\sum_i f(x_i) \tilde{w}_i}{\sum_i \tilde{w}_i} = \sum_i W_i f(x_i), \quad W_i = \frac{\tilde{w}_i}{\sum_j \tilde{w}_j}$$

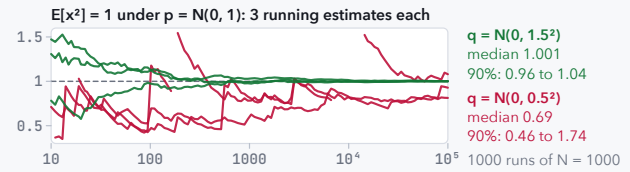
Z cancels. The ratio is biased by $O(1/N)$ but consistent. Its variance is about $\mathbb{E}_q[w^2(f - I)^2]/N$; its sd is 5.5 times below plain IS here because f barely varies where p has mass. Resampling the x_i with probabilities W_i gives approximate draws of p .

9 ESS counts what the weights are worth



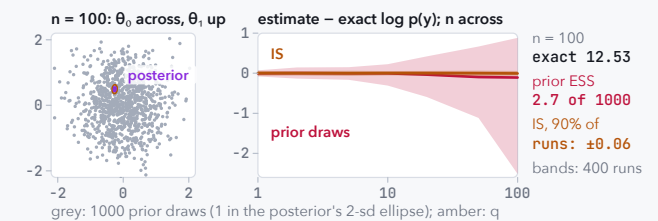
ESS = $(\sum_i w_i)^2 / \sum_i w_i^2$ runs from 1 (one weight dominates) to N (all equal) and ignores scale, so $\tilde{w}$ works. For large N , $\text{ESS}/N \rightarrow 1/\int p^2/q$: 0.72 here against a keep rate of 0.51. It sees only the weights drawn: at $\sigma_q = 0.4$ theory gives $2 \cdot 10^{-6}$, yet 100,000 draws show 0.029; the huge weights sit near $x = 4.6$, where q never goes.

10 A light-tailed q gives rare giant weights



$\mathbb{E}_q[w^2] = \int p^2/q$ is finite only if $\sigma_q^2 > \sigma_p^2/2$ (the slides' warning: $w \rightarrow \infty$ where $p \gg q$). At $\sigma_q = 0.5$ it is infinite: a run creeps, jumps at a rare tail draw (weights up to 2025) and usually reads low. Give q heavier tails than p : a Student t or a widened Gaussian.

11 The evidence: average the likelihood over the prior, or over a better q



Bayesian linear regression has $p(\mathbf{y}) = \int p(\mathbf{y} | \theta) p(\theta) d\theta = \mathcal{N}(\mathbf{y} | \mathbf{X} \mathbf{m}_0, \mathbf{X} \mathbf{S}_0 \mathbf{X}^T + \sigma^2 \mathbf{I})$; the data are the Bayesian linear regression sheet's. With prior draws, $\log p(\mathbf{y}) \approx \max \ell + \log \frac{1}{S} \sum_i e^{\ell_i - \max \ell}$, $\ell_i = \log p(\mathbf{y} | \theta_i)$ (log-sum-exp). At $n = 100$ the posterior covers 0.14% of the prior's area and the estimate reads low (mean -0.38): $\hat{Z}$ is unbiased, so $\mathbb{E}[\log \hat{Z}] \leq \log Z$. IS with $w = p(\mathbf{y} | \theta) p(\theta)/q(\theta)$ keeps 90% of runs within 0.06; with q the exact posterior every $\log w$ equals $\log p(\mathbf{y})$. The evidence notebook, on its own data (quoted): exact 12.35, 15,000 prior draws 11.20.

12 Both samplers in NumPy, in log space

```
x = rng.normal(size=n) # q = N(0, 1)
lw = log_pt(x) - log_q(x) # log w~
u = rng.random(n)
kept = x[np.log(u) < lw - logM]

w = np.exp(lw - lw.max()) # no overflow
W = w / w.sum()
est = (W * f(x)).sum() # self-normalised
ess = 1 / (W ** 2).sum()
logZ = lw.max() + np.log(w.mean())
```

Rejection: keep if $uMq(x) < \tilde{p}(x)$, rate $Z/M \cdot \text{IS}$: $\hat{I} = \sum_i W_i f(x_i)$, $\hat{Z} = \frac{1}{N} \sum_i \tilde{w}_i$, $\text{ESS} = (\sum w)^2 / \sum w^2 \cdot q$ must cover the tails of p

Source: lecture slides "Sampling methods" (the evidence frames are in the first sampling deck) and the notebooks on rejection and importance sampling and on the Monte Carlo evidence, nipunbatra.github.io/pml-teaching. Unquoted numbers: NumPy, fixed seeds. ESS, the importance curve of panel 5, the light-tail runs and the evidence comparison go beyond the slides. Corrections: the slides accept with $p/(Mq)$ for $\tilde{p}/(Mq)$; $\mathbb{E}_q[fw] = \sum_i f(x_i) w(x_i)$ lacks $1/N$, and a later step equates $\frac{1}{N} \sum f w$ with $\sum f w / \sum w$; one log-sum-exp line drops the $+$ max; the limitation figure's densities are $1/\sqrt{2}$ too low and its weight is bounded (peak 3.9); the notebook prints x at the maximiser at index 72 ($x = 1.36$), and its comments say $f = x^2$ and $w = f/q$.